

Jun Huang

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EDUCATION

• The Ohio State University

Ph.D. in Computer Science; Advisor: Prof. Feng Qin

Aug. 2021 – Aug. 2026

- **Research Focus:** System support for distributed multimodal model training and inference.

• Sun Yat-sen University

B.Sc. in Software Engineering; GPA: 4.0; Ranking: 1/334

2016 – 2020

PUBLICATIONS

1. **Jun Huang**, Zhen Zhang, Shuai Zheng, Feng Qin, and Yida Wang. **DISTMM: Accelerating Distributed Multimodal Model Training**. *NSDI*, 2024. [Link](#).
Usage: Distributed multimodal model training system that provides modality-aware parallelization, data load balancing, and topology-aware placement to improve both computation and communication efficiency. DISTMM-Pipe overcomes conventional pipeline parallelism’s batch-size limitation for contrastive learning, allowing the effective interaction batch size to scale with the number of GPUs without introducing extra idle cycles.
2. Abhinav Jangda, **Jun Huang**, Guodong Liu, Amir Hossein Nodehi Sabet, Saeed Maleki, Youshan Miao, Madanlal Musuvathi, Todd Mytkowicz, and Olli Saarikivi. **Breaking the Computation and Communication Abstraction Barrier in Distributed Machine Learning Workloads**. *ASPLOS*, 2022. [Link](#).
Usage: Generating fused kernels that extend NCCL’s existing kernels to enable arbitrary pointwise computations and reductions. Halide-inspired DSL for automatic transformation, reordering, fusing, and overlapping computation and communication in distributed machine learning workloads.
Integrated into **MSCCL**, deployed in Azure.
3. **Jun Huang**, Zhen Zhang, Xinwei Fu, Jun Wu, Yida Wang, and Feng Qin. **LOCA: Accelerating Video Generation with the Localities in Diffusion**. *On Submission*, 2026.
Usage: Accelerating video generation with a customized schedule plan that enables the colocation of incompatible video diffusion acceleration methods. Adaptively combining cross-frame and cross-step approximation methods to exploit different locality properties of the video diffusion process through a learning-based approach.

PROFESSIONAL EXPERIENCE

- **Amazon Web Services** May 2024 – Aug. 2024
Research Intern; Mentor: Zhen Zhang, Xinwei Fu, and Jun Wu.
 - **Project:** Lossy Accelerating for Video Diffusion Inference.
 - **Contribution:** Resolved a fundamental incompatibility among acceleration methods requiring conflicting execution schedules by designing a customized schedule compatible with these methods, enabling the joint exploitation of cross-frame and cross-step locality to further accelerate video diffusion inference.
- **Amazon Web Services** May 2022 – Aug. 2022; May 2023 – Aug. 2023
Research Intern; Mentors: Zhen Zhang and Shuai Zheng.
 - **Project:** Enabling Large-scale Distributed Multimodal Model Training.
 - **Contribution:** Solved the most critical scaling issue of multimodal model training’s contrastive learning, where the effective batch size remains bounded despite increasing GPU scale with pipeline parallelism. Developed **DISTMM**, the first distributed training framework specialized for multimodal models, providing improved computational efficiency and reduced communication overhead.
- **Microsoft Research Asia** Feb. 2021 – Aug. 2021
Research Intern; Mentors: Youshan Miao and Saeed Maleki.
 - **Project:** Computation–Communication Co-design for LLM Training.
 - **Contribution:** Contributed to open-source customizable collective communication library **MSCCL**. Co-developed **CoCoNet** to generate fused kernels for fine-grained communication-computation overlaps.
- **ByteDance AI Lab** Aug. 2020 – Feb. 2021
SDE Intern; Mentor: Haibin Lin.
 - **Project:** Distributed Training Framework Supporting 3D Parallelism.
 - **Contribution:** Developed 3D parallelism extension (initialization, scheduling, communication) based on open-source communication library **ps-lite** under DMLC community. Contributed to large-scale multilingual translation model training.